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DATA ENGINEERING WORD SEARCH

VOLUME 1

25 Themed Puzzles for Data Engineers,
Analysts, Students and Cloud Professionals

P Y T H O N A I R F L O W
B I G Q U E R Y D O C K E R
S N O W F L A K E C L O U D
P A R Q U E T K A F K A G I T
D A T A B R I C K S S Q L

Trivia Included • Full Solutions • For All Skill Levels

Rob Lewis

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[] Solutions	29()
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How to Use This Book

Each puzzle contains 15 data engineering words.

Words may appear:

- Horizontally
- Vertically
- Diagonally
- Forwards
- Backwards

Circle each word as you find it.

Solutions are provided at the back of the book.

Some puzzles include specialist terms from databases, cloud platforms, Python, DevOps and analytics engineering.

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Puzzle 1: Core Data Engineering Concepts

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Data engineering focuses on building reliable pipelines that move, transform and prepare data for analysis.

Z	I	D	A	T	A	B	A	S	E	C	R	D	H	T
H	Z	X	U	K	L	S	K	K	E	J	E	K	P	W
T	Q	E	L	X	K	R	V	X	S	G	N	D	O	D
L	Y	O	P	P	V	A	O	M	G	D	I	H	E	B
I	R	U	V	Y	H	V	A	P	U	D	A	T	S	B
N	E	Y	O	H	T	E	M	O	V	Y	T	I	U	Q
U	U	L	P	X	R	H	L	T	Q	L	N	F	O	G
X	Q	I	Z	T	L	C	O	S	X	A	O	Z	H	E
G	P	F	S	U	J	V	J	N	I	Z	C	O	E	L
K	G	P	A	J	M	K	W	F	L	D	F	A	R	B
I	J	P	T	W	E	X	B	F	Q	E	T	L	A	A
S	M	Y	O	E	N	I	L	E	P	I	P	I	W	T
L	X	M	B	F	P	H	C	T	A	B	Y	Z	G	I
K	B	O	R	C	H	E	S	T	R	A	T	I	O	N
R	D	F	O	M	P	P	A	M	E	H	C	S	M	T

Words:

PYTHON

TABLE

ORCHESTRATION

PIPELINE

QUERY

CONTAINER

DATABASE

ETL

CLOUD

WAREHOUSE

BATCH

LINUX

SCHEMA

STREAM

GIT

Did You Know?

Around 90% of the world's data has been created within the last few years, driven by smartphones, sensors, social media and online services.

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Puzzle 2: Databases and Warehouses

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Databases are used to store structured data, while data warehouses are optimized for analytical queries and reporting.

I	I	H	J	N	M	U	L	O	C	P	Y	D	U	T
M	R	R	O	W	A	T	A	D	A	R	E	T	H	K
G	W	B	D	O	G	N	O	M	Z	A	S	E	C	H
O	T	F	I	H	S	D	E	R	R	W	E	X	J	P
Y	C	D	T	F	M	S	Q	E	M	G	R	E	M	W
Y	U	D	Y	F	I	Y	V	O	Y	X	G	D	E	B
E	J	Y	Y	G	U	R	T	A	S	J	T	N	K	O
K	V	B	K	Y	E	E	U	O	Q	A	S	I	A	N
Y	W	E	U	S	K	U	R	R	L	C	O	P	L	T
R	V	O	L	V	Z	Q	L	A	O	Z	P	U	F	U
A	Q	Q	P	X	N	G	U	C	E	P	L	C	W	Z
M	S	X	V	D	H	I	I	L	Z	L	R	C	O	M
I	L	F	F	R	H	B	J	E	D	E	K	M	N	Y
R	D	A	T	A	B	R	I	C	K	S	A	S	S	I
P	H	G	R	V	I	E	W	M	K	M	T	U	K	F

Words:

POSTGRES

BIGQUERY

INDEX

MYSQL

SNOWFLAKE

VIEW

ORACLE

DATABRICKS

COLUMN

MONGODB

SQLSERVER

ROW

REDSHIFT

TERADATA

PRIMARYKEY

Did You Know?

The first version of Oracle was released in 1979. It was one of the first commercial relational database management systems.

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Puzzle 3: Python and Analytics Libraries

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Python is widely used in data engineering for its simplicity and rich ecosystem of libraries.

H	Y	A	C	M	Z	D	N	U	H	K	L	E	I	R
U	S	U	R	E	O	B	Q	H	V	Z	F	W	F	K
M	O	T	M	H	E	L	B	A	I	R	A	V	E	E
X	O	G	T	O	E	P	F	O	B	J	E	C	T	S
C	F	E	B	C	E	Y	K	T	S	S	F	P	U	Q
Z	M	I	K	D	Z	A	K	R	R	Y	U	M	S	L
P	O	O	L	I	D	R	E	O	A	C	N	T	S	K
S	A	D	N	A	P	R	L	P	L	N	C	P	A	T
Z	I	M	J	C	E	O	U	M	O	U	T	I	L	A
M	U	C	U	T	K	W	D	I	P	M	I	R	C	A
G	O	T	Y	K	C	D	O	S	P	P	O	C	Q	D
Z	R	P	L	I	G	U	M	L	I	Y	N	S	P	U
T	U	N	J	G	M	C	H	Z	U	F	F	X	A	G
J	F	L	H	F	E	F	M	T	M	X	E	X	D	M
H	L	W	Q	P	A	C	K	A	G	E	J	N	D	Q

Words:

PANDAS

SCRIPT

METHOD

NUMPY

MODULE

VARIABLE

POLARS

PACKAGE

IMPORT

PYARROW

FUNCTION

LOOP

JUPYTER

OBJECT

CLASS

Did You Know?

Python was named after Monty Python's Flying Circus, not the snake.

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Puzzle 4: Cloud Computing Platforms

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Cloud computing platforms provide scalable and on-demand computing resources over the internet.

W	S	B	B	E	G	C	P	U	S	L	F	I	I	Y
Q	W	Y	Y	K	F	E	R	A	W	M	V	Y	L	I
K	G	G	N	Z	N	M	A	J	J	O	X	S	V	N
L	U	D	S	J	H	I	W	D	I	J	D	A	F	S
H	A	B	N	G	P	D	W	S	S	W	F	W	O	T
J	D	E	E	K	R	O	W	T	E	N	S	S	E	A
C	B	T	E	R	U	Z	A	I	B	U	T	V	W	N
I	M	U	W	A	N	C	L	O	U	D	R	U	N	C
R	A	P	Q	Y	T	E	C	W	C	D	A	U	F	E
B	L	M	Z	D	Z	S	T	D	H	Q	S	R	A	R
A	H	O	I	O	R	Q	P	E	K	Z	M	J	V	F
F	O	C	S	F	U	N	D	V	S	R	S	R	D	F
Y	Z	D	F	T	S	T	O	R	A	G	E	Y	B	M
E	Q	M	L	W	S	T	E	K	C	U	B	W	N	W
Q	Q	R	E	V	R	E	S	F	R	W	C	N	O	M

Words:

AZURE

NETWORK

FABRIC

AWS

SERVER

LAMBDA

GCP

INSTANCE

CLOUDRUN

COMPUTE

BUCKET

VMWARE

STORAGE

IAAS

KUBERNETES

Did You Know?

Amazon Web Services launched in 2006 with just a handful of services.

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Puzzle 5: Data Formats and Storage

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Data can be stored in various formats, each with its own advantages for storage efficiency and query performance.

W	R	E	C	O	R	D	H	D	U	O	C	V	Q	N
D	E	C	O	D	E	X	M	L	K	F	F	E	D	E
K	T	O	S	F	A	O	P	C	M	G	V	D	X	F
F	E	F	E	D	O	J	S	O	N	R	N	O	L	B
T	U	Y	R	S	L	R	G	Z	U	E	C	C	O	O
Q	Q	D	I	A	M	N	M	R	K	B	D	N	W	F
S	R	O	A	U	A	O	R	A	E	E	V	E	I	J
E	A	R	L	U	Q	I	L	H	T	C	N	J	H	N
A	P	V	I	V	B	T	R	Q	A	I	J	V	U	Z
S	L	A	Z	T	N	I	C	O	M	P	R	E	S	S
S	D	W	E	B	P	T	U	T	F	R	D	U	E	E
X	H	A	P	K	D	R	J	C	A	P	E	W	M	X
C	W	U	K	O	X	A	B	S	X	C	L	U	B	A
H	M	H	U	M	R	P	V	S	C	C	T	R	Y	G
Z	N	K	K	F	I	E	L	D	M	Z	A	R	I	D

Words:

PARQUET

DELTA

SERIALIZE

AVRO

ICEBERG

RECORD

JSON

FORMAT

FIELD

CSV

PARTITION

ENCODE

XML

COMPRESS

DECODE

Did You Know?

A Parquet file can often be significantly smaller than the same data stored as CSV.

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Puzzle 6: DevOps and Deployment

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DevOps practices aim to streamline the software development and deployment process through automation and collaboration.



Words:

TERRAFORM

GITHUB

VERSION

ANSIBLE

BITBUCKET

MERGE

DOCKER

DEPLOY

COMMIT

PODMAN

BUILD

BRANCH

JENKINS

RELEASE

PULLREQUEST

Did You Know?

The term 'DevOps' was popularised in 2009 by Patrick Debois.

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Puzzle 7: Data Quality and Governance

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Data quality and governance ensure that data is accurate, consistent, and compliant with organizational policies.

S	L	N	A	S	E	T	I	D	U	A	X	Q	M	N
R	P	O	L	C	K	Y	V	G	N	I	S	S	I	M
O	M	I	Q	O	W	F	P	W	A	R	N	I	N	G
R	U	T	L	N	O	N	F	M	K	H	O	V	U	K
R	S	A	K	T	R	F	V	H	T	F	A	D	V	S
E	K	D	I	R	J	U	M	F	L	T	T	D	G	N
J	C	I	N	O	C	D	L	O	H	S	E	R	H	T
M	E	L	E	L	G	R	Z	E	G	W	N	W	G	V
O	H	A	B	X	R	A	I	Y	C	Q	Q	D	H	F
N	C	V	U	G	J	W	L	D	C	F	U	Y	P	Y
I	C	G	G	X	T	Y	V	N	U	S	N	O	B	D
T	X	Q	U	A	L	I	T	Y	L	A	M	O	N	A
O	P	R	O	F	I	L	E	S	V	G	R	K	F	Q
R	N	L	X	D	U	P	L	I	C	A	T	E	W	I
P	J	J	D	M	H	A	T	A	D	A	T	E	M	M

Words:

QUALITY
VALIDATION
PROFILE
DUPLICATE
MISSING

ANOMALY
CHECKSUM
METADATA
MONITOR
AUDIT

ERROR
WARNING
THRESHOLD
CONTROL
RULE

Did You Know?

Bad data quality costs businesses billions of dollars every year.

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Puzzle 8: Workflow Orchestration

{ }

Workflow orchestration involves managing and automating complex workflows across multiple systems and services.

C	U	I	L	J	P	T	F	N	A	S	Q	K	K	F
Q	C	E	J	A	E	F	E	N	X	X	M	V	S	L
R	A	P	C	T	U	Q	T	F	O	Q	J	V	A	S
O	I	S	R	X	E	U	A	D	J	F	E	T	T	W
T	R	P	E	A	U	E	R	K	R	I	L	R	Y	O
A	F	R	T	U	Q	R	T	A	E	D	U	I	R	L
R	L	E	S	B	H	V	S	L	N	R	D	G	T	F
E	O	F	G	Q	W	V	E	M	N	R	E	G	E	K
P	W	E	A	L	J	V	H	M	U	O	H	E	R	R
O	E	C	D	V	O	X	C	A	R	S	C	R	X	O
T	R	T	V	Z	P	P	R	X	G	N	S	N	N	W
E	X	E	C	U	T	I	O	N	E	E	M	E	R	I
H	N	S	X	T	R	L	H	U	D	S	B	O	Z	F
Y	C	N	E	D	N	E	P	E	D	U	E	U	B	I
F	D	F	L	L	E	K	N	N	C	E	W	J	R	Y

Words:

AIRFLOW

WORKFLOW

OPERATOR

DAGSTER

TASK

SENSOR

PREFECT

RETRY

QUEUE

SCHEDULE

DEPENDENCY

RUNNER

TRIGGER

EXECUTION

ORCHESTRATE

Did You Know?

Apache Airflow was originally developed at Airbnb.

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Puzzle 9: Data Modelling

{ }

Data modelling is the process of creating a conceptual representation of data and its relationships.

D	U	M	K	K	R	E	P	Q	A	E	F	K	S	U
Q	N	E	U	U	A	T	I	B	V	T	B	A	A	F
K	I	A	B	C	T	A	I	G	F	U	U	W	N	E
U	Q	S	R	I	S	G	G	D	A	B	B	F	A	A
I	U	U	Q	N	Y	O	J	I	C	I	I	T	T	Z
I	E	R	I	O	H	R	C	M	T	R	G	R	U	E
D	K	E	U	R	C	R	B	E	T	T	C	A	R	S
B	E	Z	X	M	R	U	I	N	Y	T	J	M	A	U
I	Y	O	A	A	A	S	W	S	Q	A	U	H	L	O
Y	D	T	X	L	R	K	J	I	D	B	D	G	K	H
K	Z	L	A	I	E	X	B	O	N	G	Q	M	E	E
T	H	S	Z	Z	I	P	E	N	T	I	T	Y	Y	K
X	U	T	I	E	H	L	E	D	O	M	A	X	W	A
R	A	T	L	V	W	H	W	K	M	L	Q	Z	U	L
D	L	H	V	R	E	L	A	T	I	O	N	P	E	C

Words:

STAR	NATURALKEY	ENTITY
UNIQUEKEY	ATTRIBUTE	RELATION
DIMENSION	MEASURE	MART
FACT	HIERARCHY	LAKEHOUSE
SURROGATE	MODEL	NORMALIZE

Did You Know?

Ralph Kimball and Bill Inmon helped define many modern data warehousing practices.

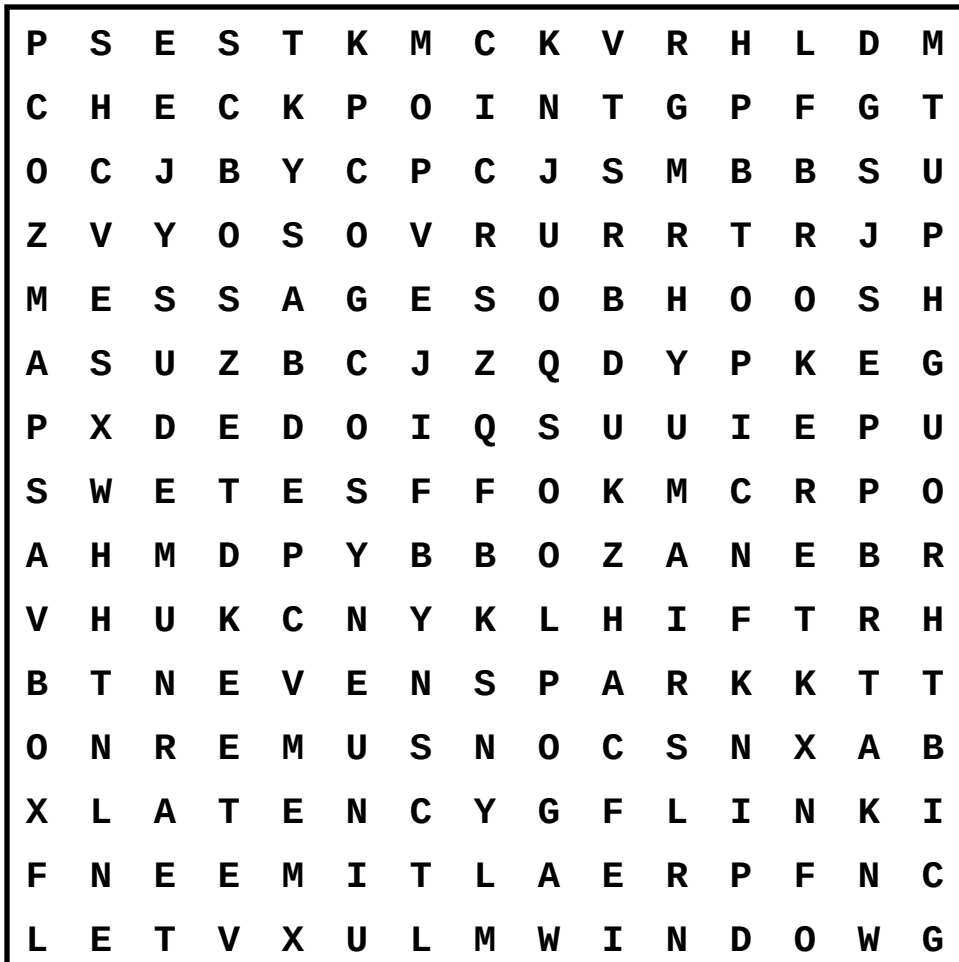
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Puzzle 10: Streaming and Event Processing { }

Streaming and event processing involve handling and analyzing data in real-time as it is generated.



Words:

MESSAGES

KAFKA

SPARK

FLINK

EVENT

TOPIC

BROKER

CONSUMER

PRODUCER

OFFSET

WINDOW

LATENCY

THROUGHPUT

REALTIME

CHECKPOINT

Did You Know?

Apache Kafka was originally developed at LinkedIn and later became an open-source project.

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Puzzle 11: SQL and Query Logic

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SQL (Structured Query Language) is used to manage and manipulate relational databases.

A	H	U	N	I	O	N	Y	S	D	Y	Z	Q	U	N
N	A	Q	T	E	S	F	B	F	I	L	T	E	R	G
X	V	F	G	X	I	E	R	Z	E	L	V	R	R	I
S	I	K	H	Y	A	C	E	I	Y	T	S	E	D	V
L	N	Z	W	X	O	E	D	C	S	P	U	T	H	E
K	G	U	O	Z	X	T	R	H	E	D	B	A	Q	T
G	W	J	U	I	V	A	O	J	L	L	Q	G	N	E
E	R	H	R	B	Y	D	E	H	E	F	U	E	E	L
G	C	O	E	N	G	P	T	Z	C	L	E	R	Z	E
M	J	N	U	R	L	U	R	N	T	J	R	G	K	D
I	X	C	J	P	E	P	E	S	H	Z	Y	G	N	D
O	J	H	D	A	B	U	S	N	S	C	J	A	I	A
D	I	I	Q	R	P	Y	N	F	K	O	H	Z	O	W
E	B	E	T	C	R	S	I	K	P	D	O	Z	J	I
G	E	O	E	G	U	Z	Y	J	P	S	O	R	T	J

Words:

SELECT

JOIN

SUBQUERY

INSERT

GROUPBY

CTE

UPDATE

ORDERBY

FILTER

DELETE

HAVING

SORT

WHERE

UNION

AGGREGATE

Did You Know?

SQL was originally developed by IBM researchers in the 1970s.

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Puzzle 12: Big Data Processing

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Big data processing involves handling and analyzing large volumes of data that cannot be managed with traditional data processing tools.

D	G	W	E	R	E	L	F	F	U	H	S	D	V	X
W	I	G	D	Z	R	E	T	S	U	L	C	H	K	Y
E	G	F	O	L	D	R	I	V	E	R	L	D	E	G
K	X	Y	N	E	E	T	U	B	I	R	T	S	I	D
A	F	I	B	E	J	E	W	U	K	K	C	F	Y	J
D	L	T	C	X	X	B	S	E	D	H	Q	Q	F	S
P	I	G	U	E	L	R	M	C	Q	A	W	D	U	E
H	X	Y	P	C	P	B	G	U	S	D	O	I	H	R
I	X	R	E	U	Z	R	D	D	E	O	R	N	L	O
V	B	O	Z	T	X	U	E	E	O	O	K	N	E	C
E	C	M	T	O	A	K	W	R	P	P	E	Z	P	P
P	Z	E	Z	R	W	W	M	P	P	X	R	P	E	X
S	U	M	D	Z	N	P	E	A	A	M	W	G	L	G
C	S	Y	F	V	P	S	W	M	A	A	Q	Z	V	X
Z	P	D	A	T	A	F	R	A	M	E	F	M	J	G

Words:

HADOOP

EXECUTOR

WORKER

HIVE

DRIVER

NODE

PIG

SHUFFLE

MEMORY

MAPREDUCE

RDD

CORES

CLUSTER

DATAFRAME

DISTRIBUTE

Did You Know?

Hadoop was inspired by papers published by Google.

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Puzzle 13: Data Lakes

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Data lakes are centralized repositories that allow you to store all your structured and unstructured data at any scale.

O	U	J	L	H	G	M	P	K	Z	F	Q	V	Z	O
B	L	O	B	I	N	G	E	S	T	B	P	L	D	K
M	D	V	G	Q	I	G	M	E	U	E	U	K	O	I
J	X	S	B	F	D	R	E	T	E	N	T	I	O	N
N	L	C	B	T	N	K	Y	K	B	P	A	T	H	J
C	U	O	E	G	A	E	N	I	L	S	R	Q	Z	H
M	R	W	A	P	L	A	R	C	H	I	V	E	T	Y
Q	E	Y	E	R	O	T	S	T	C	E	J	B	O	N
X	D	J	S	Z	K	W	X	X	D	E	R	P	D	G
T	L	P	U	T	R	C	U	E	N	A	B	U	E	O
Q	O	D	M	E	W	S	T	O	V	O	R	A	X	L
T	F	X	V	J	Y	A	Z	D	T	W	O	W	P	A
A	I	L	Z	N	R	W	N	L	J	O	N	F	W	T
E	I	L	R	U	A	Z	E	O	Z	R	Z	Z	J	A
S	T	D	C	R	N	J	Y	G	K	A	E	E	Y	C

Words:

RAWZONE

CURATED

ARCHIVE

BRONZE

LANDING

BLOB

SILVER

CATALOG

OBJECTSTORE

GOLD

LINEAGE

FOLDER

INGEST

RETENTION

PATH

Did You Know?

The term 'data lake' was coined by James Dixon in 2010.

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Puzzle 14: Analytics Engineering

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Analytics engineering involves building and maintaining the infrastructure and processes needed to support data analysis and reporting.

S	X	A	U	Z	O	M	R	O	F	S	N	A	R	T
N	V	Q	Q	X	V	H	D	H	M	E	T	R	I	C
A	G	W	I	D	C	I	T	N	A	M	E	S	A	P
P	S	O	S	O	U	R	C	E	Y	P	H	R	E	X
S	B	R	T	C	K	N	C	U	F	Q	H	Q	Z	W
H	P	C	E	U	E	G	E	B	G	P	G	Q	L	S
O	P	A	S	M	R	F	F	Q	E	L	N	F	O	T
T	E	M	T	E	E	G	D	Y	G	X	I	N	H	D
Z	P	J	U	N	F	O	M	I	Q	Z	G	F	L	M
O	W	L	T	T	B	I	H	Q	G	T	A	Y	L	H
E	R	U	S	O	P	X	E	U	O	B	T	U	J	F
Q	Y	A	M	L	F	N	O	M	D	D	S	E	E	D
N	G	B	N	D	W	M	R	V	C	L	C	I	O	Y
M	A	T	E	R	I	A	L	I	Z	E	Z	A	B	Q
Y	E	G	U	D	I	Y	K	F	Q	O	G	F	Y	M

Words:

DBT

TEST

MATERIALIZER

SEED

REF

EXPOSURE

SOURCE

YAML

DOCUMENT

SNAPSHOT

STAGING

SEMANTIC

MACRO

TRANSFORM

METRIC

Did You Know?

Analytics engineering is a relatively new role that bridges the gap between data engineering and data analysis.

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Puzzle 15: APIs and Data Exchange

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APIs (Application Programming Interfaces) enable different software systems to communicate and exchange data.

X	W	K	Y	V	T	H	O	N	H	S	J	Z	E	H
K	T	I	M	I	L	E	T	A	R	K	T	A	S	M
P	R	H	P	H	E	U	X	L	F	L	M	I	N	S
D	N	E	W	S	A	P	E	V	T	W	D	K	O	S
W	A	A	S	P	R	K	O	O	H	B	E	W	P	E
D	L	D	S	T	A	P	A	Y	L	O	A	D	S	L
B	Q	E	R	P	D	G	J	B	R	X	Z	N	E	R
S	H	R	E	T	S	L	I	G	U	F	K	W	R	E
I	P	R	Q	Z	V	C	P	N	O	B	A	U	P	V
C	A	V	U	N	E	K	O	T	A	Y	X	N	A	R
K	R	M	E	I	B	D	U	A	N	T	S	E	U	E
F	G	L	S	O	Q	Q	N	D	P	X	I	T	T	S
J	P	V	T	C	A	M	U	I	U	I	J	O	H	I
T	Y	T	T	N	I	O	P	D	N	E	M	J	N	S
B	R	C	L	I	E	N	T	V	G	N	P	F	B	H

Words:

API

REST

GRAPHQL

ENDPOINT

REQUEST

RESPONSE

PAYLOAD

HEADER

TOKEN

AUTH

PAGINATION

WEBHOOK

RATELIMIT

CLIENT

SERVERLESS

Did You Know?

The first web API was introduced by Salesforce in 2000.

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Puzzle 16: Security and Access

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Security and access control are critical in data engineering to protect sensitive data and ensure that only authorized users can access it.

L	C	B	T	E	H	S	E	C	R	E	T	D	M	A
P	B	E	L	R	O	Y	H	W	A	V	T	A	U	Y
L	R	V	R	V	E	K	V	G	C	T	I	Y	W	N
F	Z	I	T	T	D	D	N	W	T	N	E	K	O	K
X	I	W	N	V	I	I	A	L	S	K	U	I	J	P
M	E	R	E	C	K	F	U	C	E	I	S	G	N	U
S	Z	I	E	S	I	A	I	T	T	S	Y	O	W	B
I	I	K	A	W	V	P	A	C	I	I	I	J	B	L
I	N	M	Z	Y	A	V	A	M	A	T	O	T	U	I
W	E	U	E	O	I	L	R	L	P	T	Z	N	J	C
E	K	K	M	R	Q	E	L	Y	Q	R	E	B	E	K
L	O	K	P	A	P	K	R	C	S	P	V	B	A	E
O	T	M	D	K	H	C	L	R	D	D	V	O	E	Y
R	N	L	Z	R	N	Y	C	I	L	O	P	M	Z	C
J	W	T	E	E	L	M	S	J	U	T	R	A	D	O

Words:

ENCRYPTION

KEYVAULT

TOKENIZE

IAM

FIREWALL

MASKING

ROLE

PRIVATEKEY

REDACTION

POLICY

PUBLICKEY

PERMISSION

SECRET

CERTIFICATE

PRINCIPAL

Did You Know?

The strongest encryption algorithms would take modern computers an impractically long time to brute force.

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Puzzle 17: Monitoring and Logging

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Monitoring and logging involve tracking and recording system activities to ensure proper operation and facilitate troubleshooting.

G	W	Z	Q	B	M	U	P	T	I	M	E	F	A	N
O	A	A	Y	G	T	E	C	A	R	T	H	T	L	K
D	V	A	L	S	Q	V	N	V	O	N	C	J	D	C
H	F	T	O	X	L	O	G	G	I	N	G	M	A	E
C	M	T	E	L	E	M	E	T	R	Y	O	N	S	H
T	U	E	P	T	Q	Q	K	R	K	J	U	I	H	C
A	L	C	T	U	F	Y	E	T	S	S	A	H	B	H
W	U	G	M	R	A	X	R	O	K	T	L	E	O	T
B	S	I	N	C	I	D	E	N	T	A	E	H	A	L
W	B	Z	T	X	L	C	S	S	J	T	R	Q	R	A
K	U	X	G	J	U	Q	S	I	X	U	T	C	D	E
G	L	O	Y	A	R	U	A	T	G	S	C	V	G	H
C	J	O	I	E	E	F	U	Q	O	N	D	T	J	I
Y	O	B	S	E	R	V	E	U	U	R	A	T	N	O
A	M	Z	K	C	I	I	E	J	E	T	E	L	Q	J

Words:

LOGGING

OBSERVE

HEALTHCHECK

TRACE

UPTIME

TELEMETRY

ALERT

FAILURE

WATCHDOG

METRICSTORE

INCIDENT

STATUS

DASHBOARD

SLA

SIGNAL

Did You Know?

The Mars Climate Orbiter was lost because of a units mismatch between systems.

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Puzzle 18: Testing Data Pipelines

{ }

Testing data pipelines involves verifying that data flows correctly through each stage of the pipeline and that the output meets the expected quality standards.

V	I	A	A	S	S	E	R	T	R	K	C	K	H	L
S	H	N	U	X	C	L	N	Y	K	S	O	U	I	D
M	G	I	V	T	I	R	O	D	X	E	V	W	H	S
O	C	N	T	A	N	Q	I	B	I	C	E	T	S	D
K	V	T	L	L	L	K	S	A	V	B	R	U	Z	Z
E	K	E	U	V	G	C	S	S	D	M	A	N	P	H
T	C	G	S	Y	E	Q	E	E	E	E	G	I	G	A
E	O	R	E	A	U	M	R	L	T	R	E	T	V	C
S	M	A	R	L	K	U	G	I	C	A	R	T	F	T
T	F	T	U	I	D	K	E	N	E	P	U	E	L	U
V	V	I	F	Z	Y	V	R	E	P	M	T	S	I	A
D	N	O	O	Z	M	I	F	V	X	O	X	T	T	L
X	B	N	E	X	D	V	H	Z	E	C	I	A	C	I
L	P	Y	T	E	S	T	A	E	D	A	F	E	B	I
X	X	H	Y	S	C	E	N	A	R	I	O	W	W	P

Words:

UNITTEST

EXPECTED

COVERAGE

PYTEST

ACTUAL

SCENARIO

FIXTURE

REGRESSION

BASELINE

MOCK

SMOKETEST

COMPARE

ASSERT

INTEGRATION

RESULT

Did You Know?

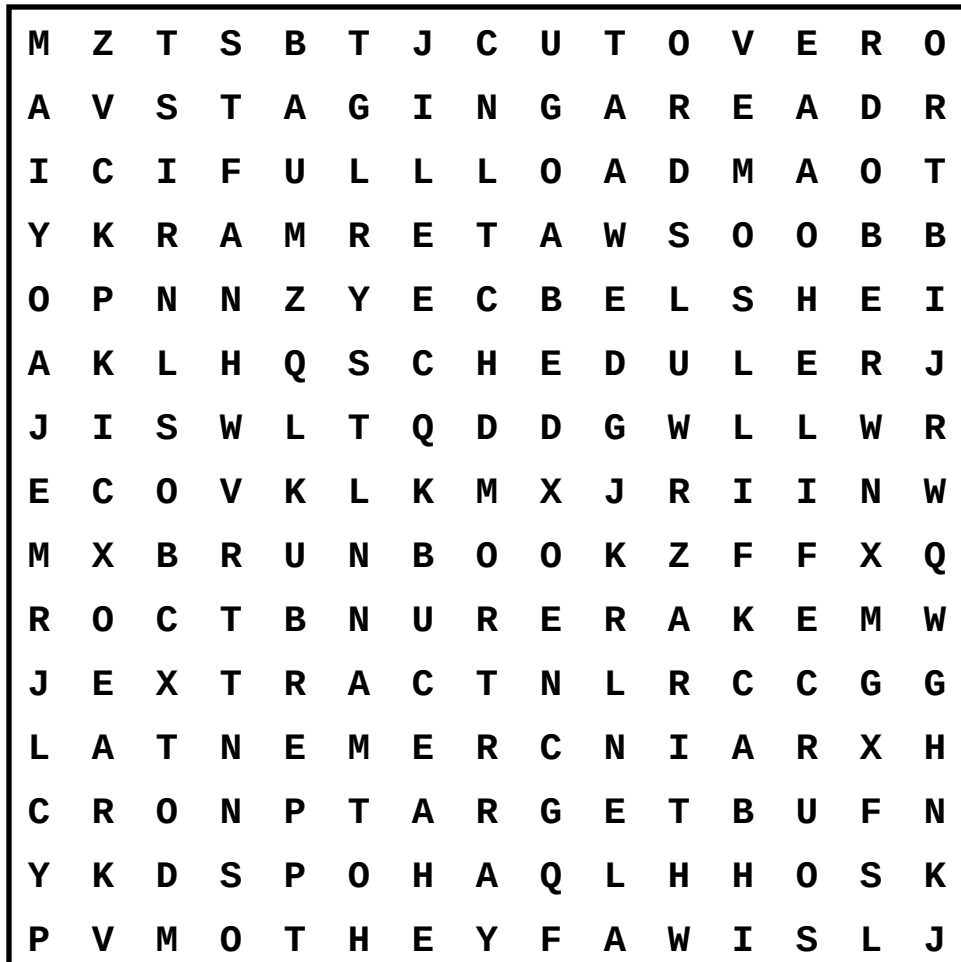
NASA software projects often devote more effort to testing than coding.

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Puzzle 19: Batch Processing

Batch processing involves executing a series of jobs in a queued manner without manual intervention.



Words:

JOB	LOAD	INCREMENTAL
SCHEDULER	EXTRACT	FULLLOAD
CRON	SOURCEFILE	WATERMARK
BACKFILL	TARGET	CUTOVER
RERUN	STAGINGAREA	RUNBOOK

Did You Know?

Batch processing has been around since the earliest mainframe computers.

Puzzle 20: Data Architecture

Data architecture is the design and implementation of the overall structure and organization of an organization's data assets.



Words:

MEDALLION

BUS

BLUEPRINT

HUB

VAULT

LAYER

SPOKE

SATELLITE

PLATFORM

DOMAIN

LINK

SERVICE

MESH

HISTORY

CAPABILITY

Did You Know?

Some enterprise data architectures contain thousands of interconnected systems.

</>

Puzzle 21: Cloud Data Services

{ }

Cloud data services provide managed solutions for storing, processing, and analyzing data in the cloud.



Words:

ATHENA

PUBSUB

DYNAMODB

GLUE

KINESIS

BIGTABLE

EMR

FIREHOSE

SPANNER

SYNAPSE

EVENTHUB

DATAPROC

DATAFLOW

COSMOSDB

DATAFORM

Did You Know?

Google processes billions of search queries every day.

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()

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Puzzle 22: Performance Tuning

{ }

Performance tuning involves optimizing database and application performance to improve response times and throughput.

J	D	X	F	S	N	C	B	E	P	L	A	N	O	F
X	B	O	T	T	L	E	N	E	C	K	C	P	P	G
S	X	G	S	K	V	W	B	R	P	E	X	A	L	S
C	T	N	E	B	K	H	R	M	O	H	G	R	G	V
I	U	I	E	Y	R	T	O	Z	P	C	W	A	Y	M
T	U	R	K	O	D	E	A	N	T	A	S	L	L	L
S	D	E	Y	U	Q	X	D	R	I	C	S	L	V	Z
I	V	T	T	I	L	P	C	J	M	E	G	E	C	N
T	D	S	Z	M	A	L	A	O	I	Q	N	L	Y	I
A	Z	U	U	U	M	A	S	Q	Z	L	I	B	N	O
T	S	L	Z	U	E	I	T	X	E	Z	N	M	N	J
S	E	C	F	C	D	N	N	Y	L	R	U	X	O	H
O	T	U	Y	A	T	S	O	C	S	H	R	J	T	S
W	V	L	V	V	W	X	L	S	J	O	P	S	H	A
B	P	G	N	A	C	S	V	T	L	C	V	F	Y	H

Words:

OPTIMIZE

VACUUM

HASHJOIN

CACHE

EXPLAIN

BROADCAST

PRUNING

PLAN

PARALLEL

CLUSTERING

SCAN

BOTTLENECK

STATISTICS

SEEK

COST

Did You Know?

A well-designed index can turn a query from minutes into milliseconds.

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()

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Puzzle 23: File Handling

{ }

File handling involves reading, writing, and managing files in various formats for data processing and storage.



Words:

FILENAME
DIRECTORY
EXTENSION
DELIMITER
ENCODING

SCHEMAFILE
MANIFEST
CHUNK
SPLIT
MERGEFILE

UPLOAD
DOWNLOAD
ARCHIVING
ROTATION
LOCKFILE

Did You Know?

CSV was never formally standardised for much of its history.

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Puzzle 24: Business Intelligence

Business intelligence involves the strategies and technologies used to gather, analyze, and present business data for decision-making.



Words:

REPORT

CHART

VISUAL

TABLEAU

POWERBI

LOOKER

FILTERING

DRILLDOWN

KPI

MEASURED

SUMMARY

EXPORT

REFRESH

PUBLISH

INSIGHT

Did You Know?

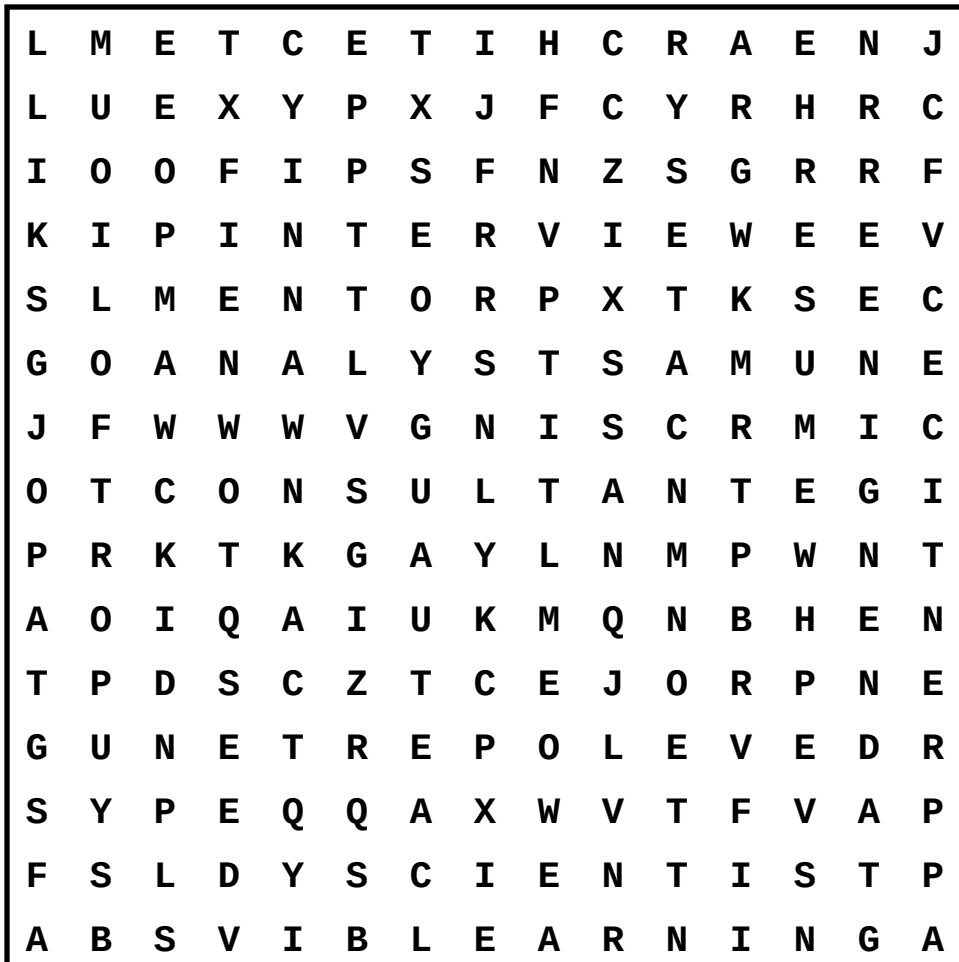
The first spreadsheet program appeared decades before modern BI tools.

</>

Puzzle 25: Data Careers

{ }

Data careers encompass a wide range of roles and specializations in the field of data engineering, analytics, and science.



Words:

ANALYST

CONSULTANT

INTERVIEW

ENGINEER

SPECIALIST

RESUME

ARCHITECT

APPRENTICE

PROJECT

DEVELOPER

MENTOR

SKILL

SCIENTIST

PORTFOLIO

LEARNING

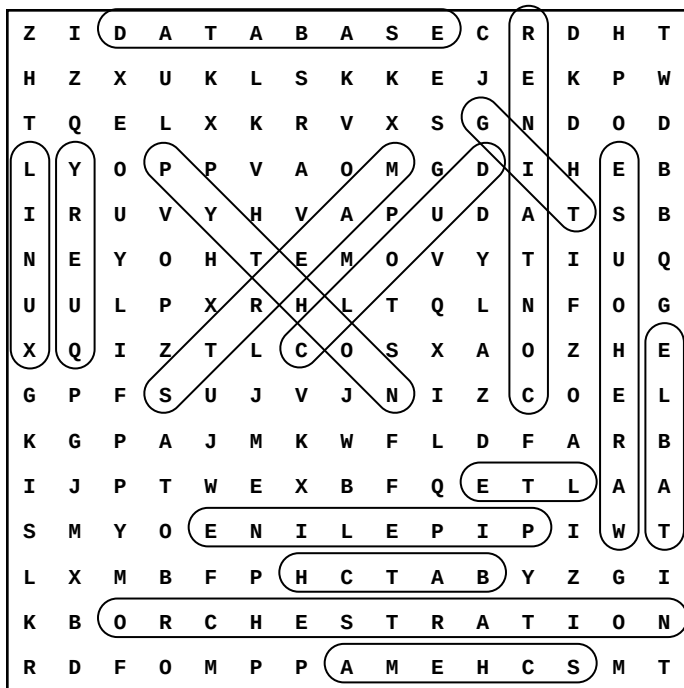
Did You Know?

Demand for data engineers has grown dramatically over the last decade.

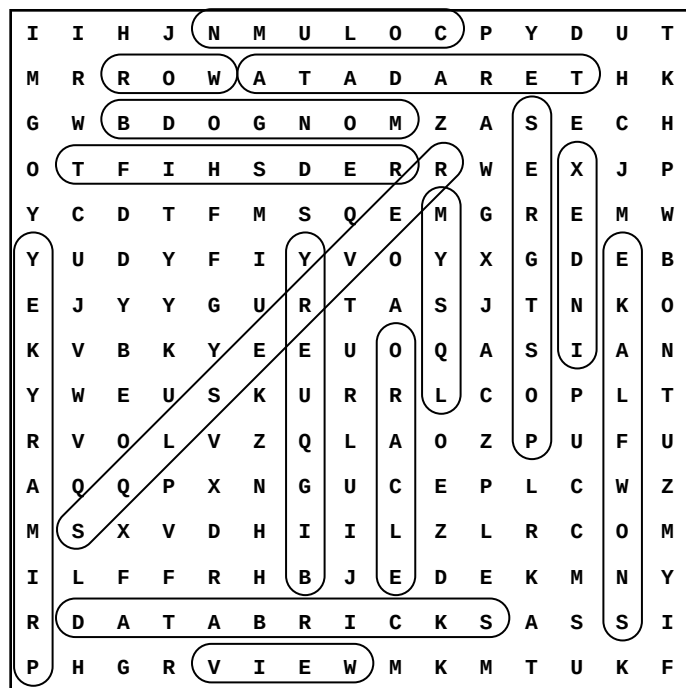
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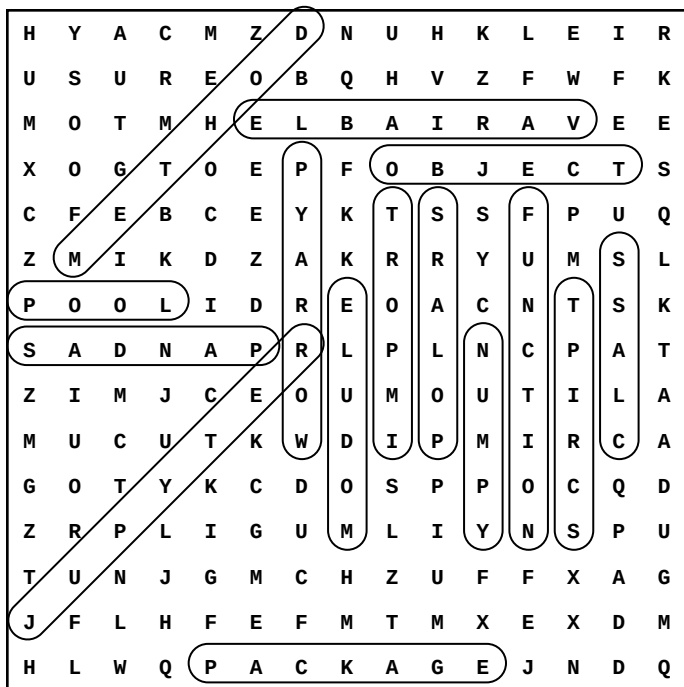
Solution 1



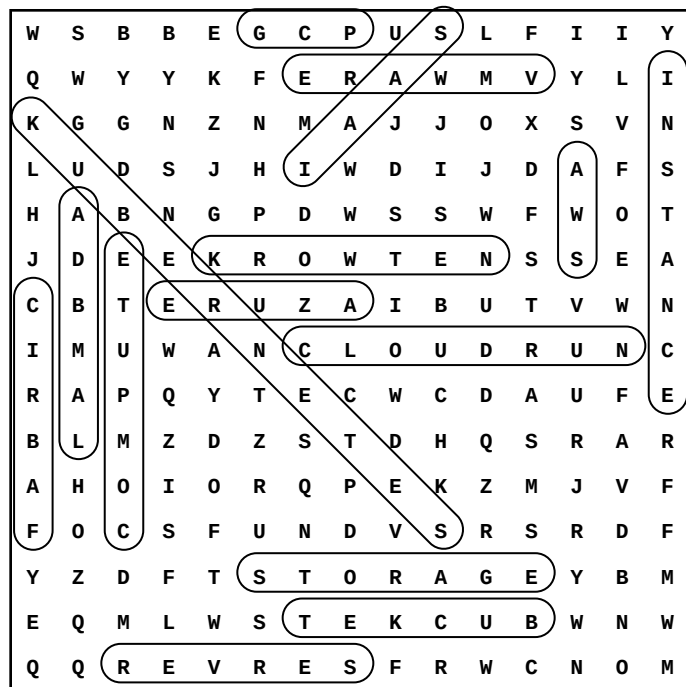
Solution 2



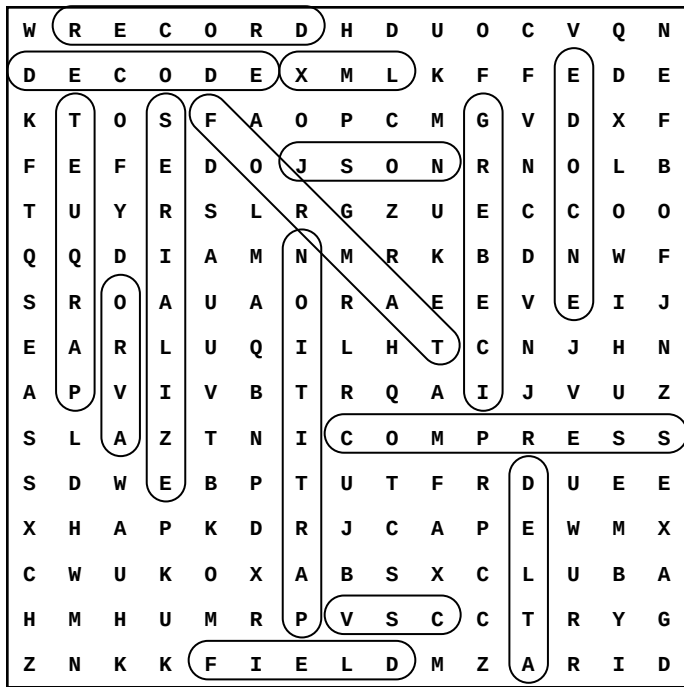
Solution 3



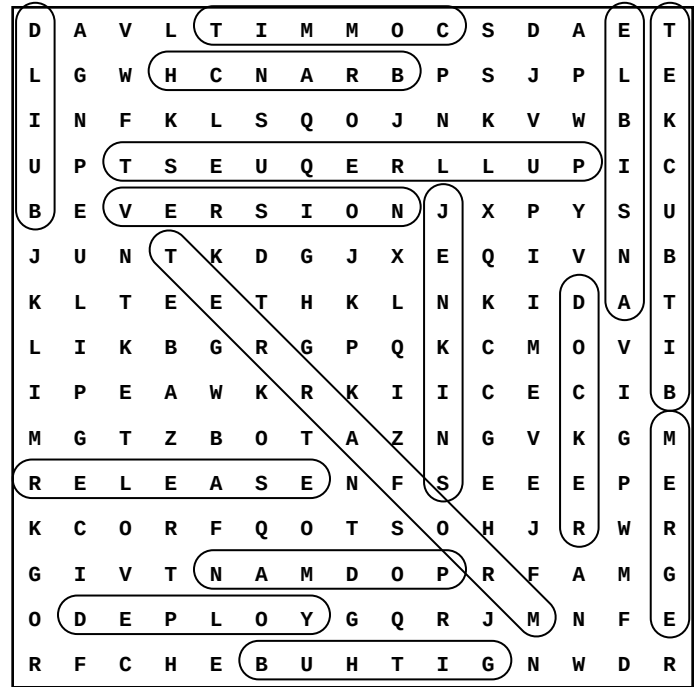
Solution 4



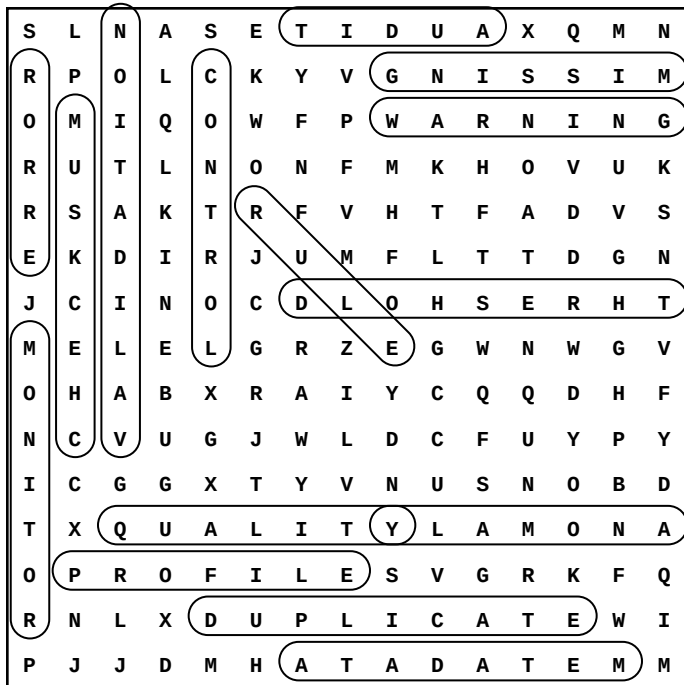
Solution 5



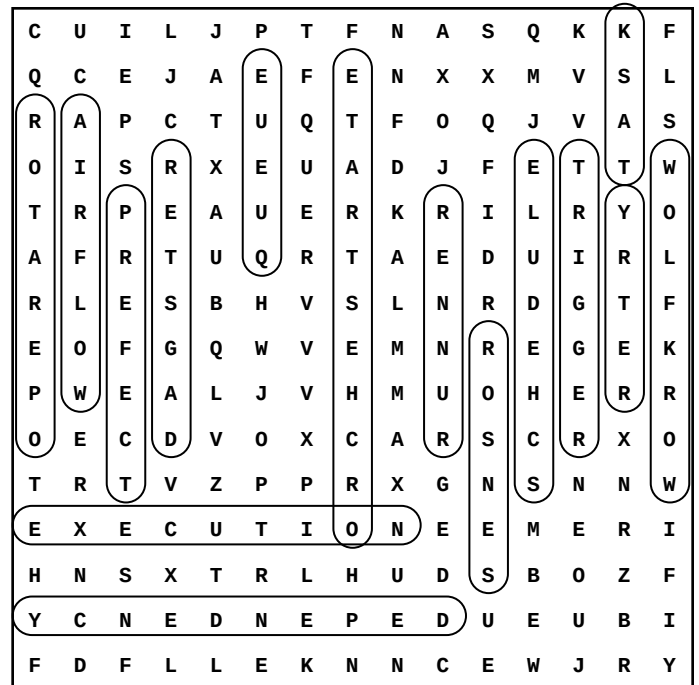
Solution 6



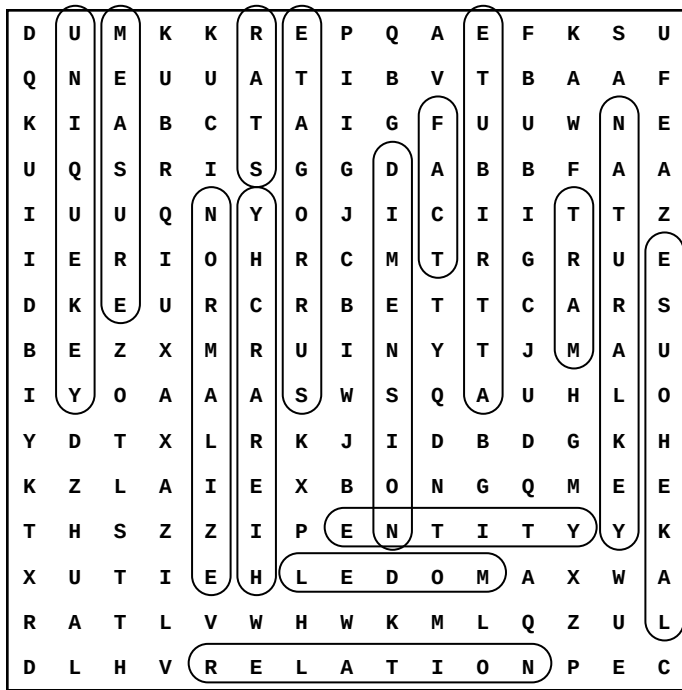
Solution 7



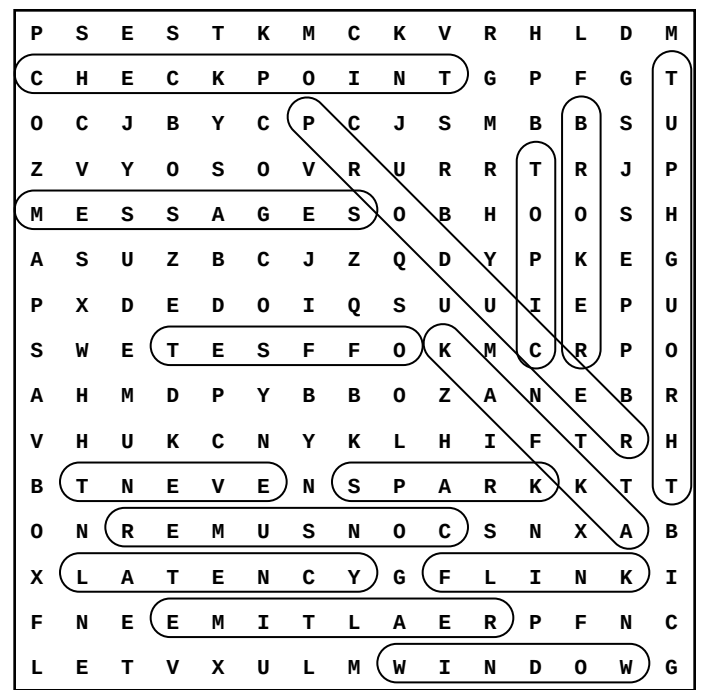
Solution 8



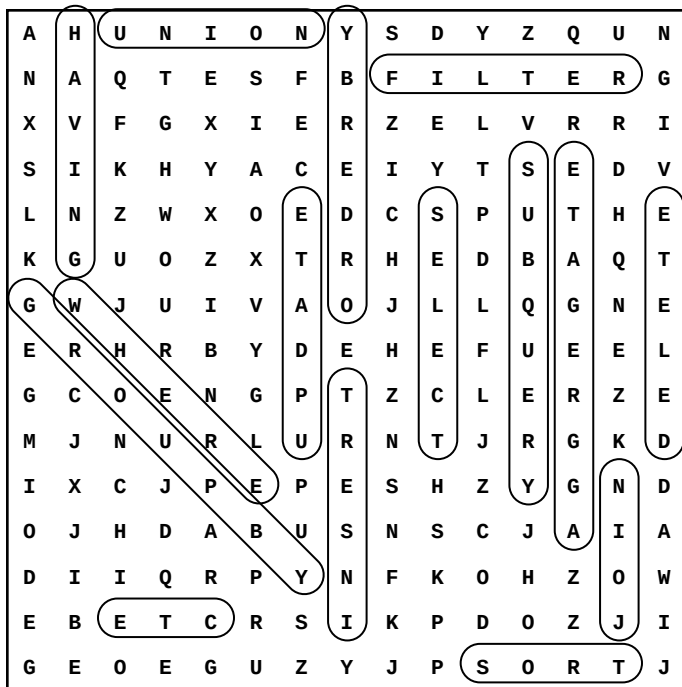
Solution 9



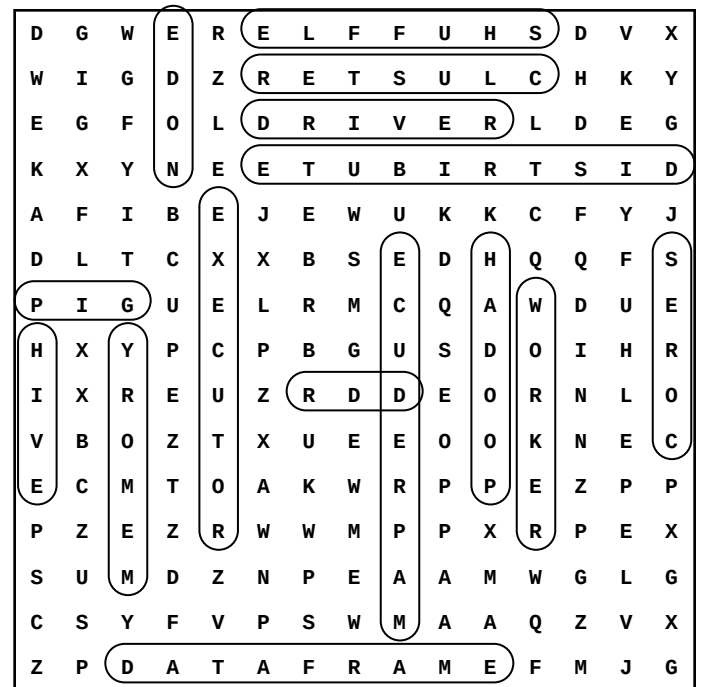
Solution 10



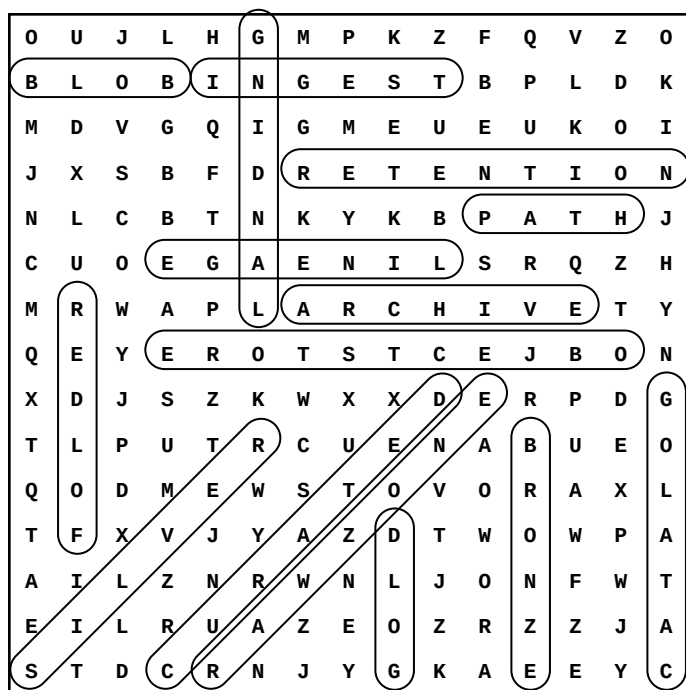
Solution 11



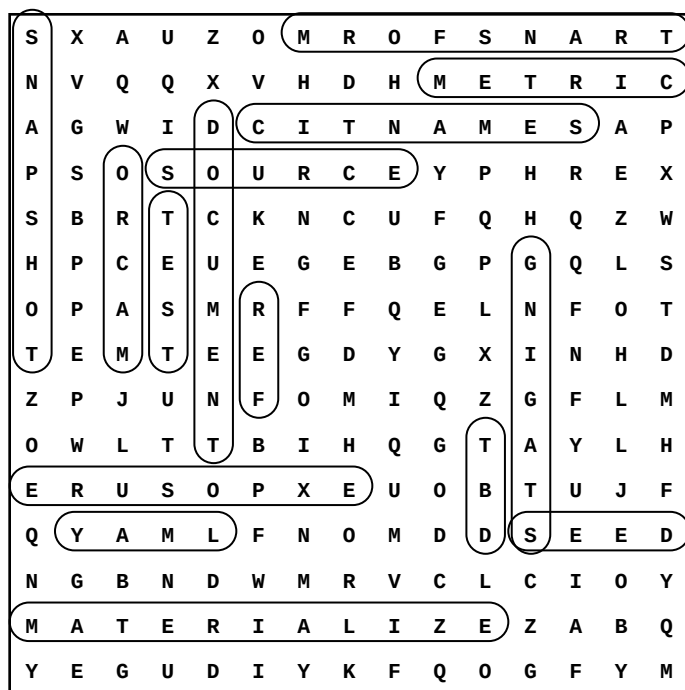
Solution 12



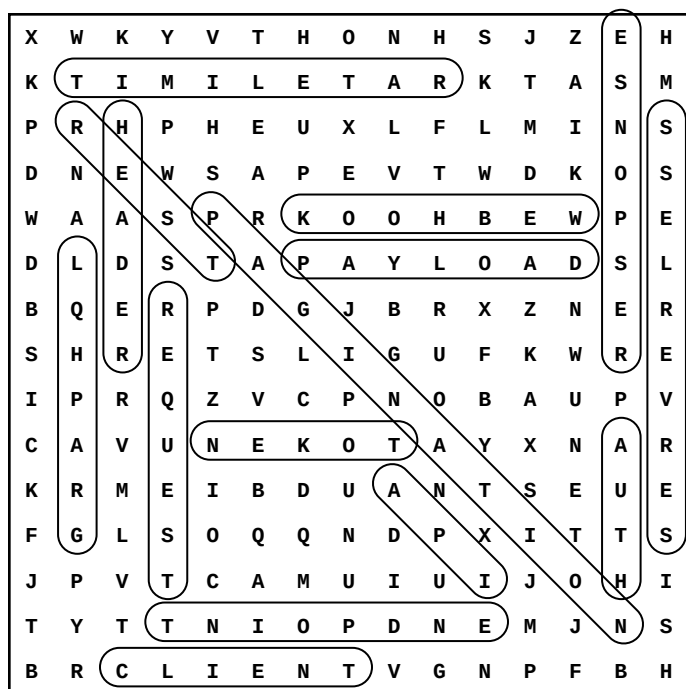
Solution 13



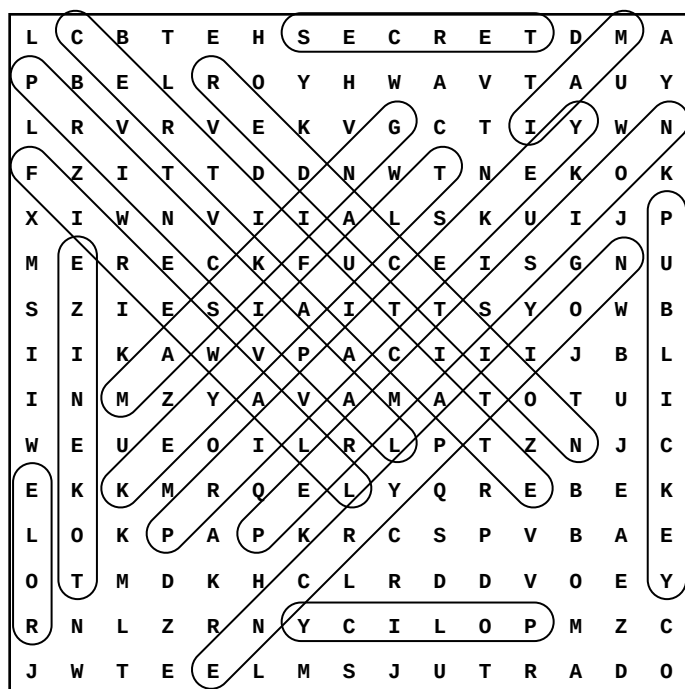
Solution 14



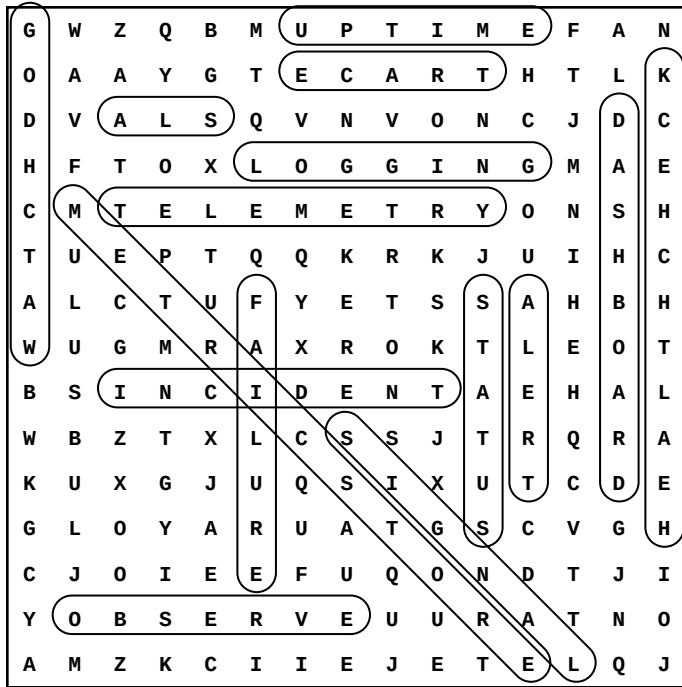
Solution 15



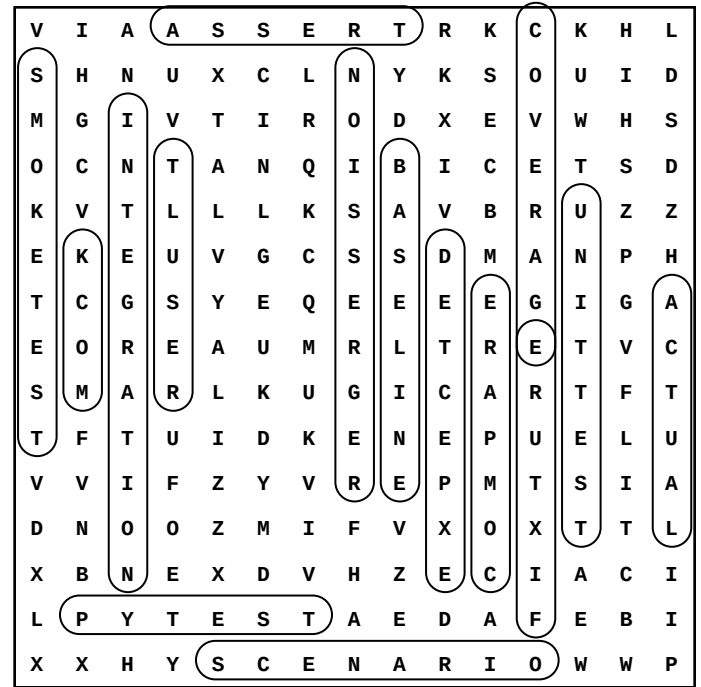
Solution 16



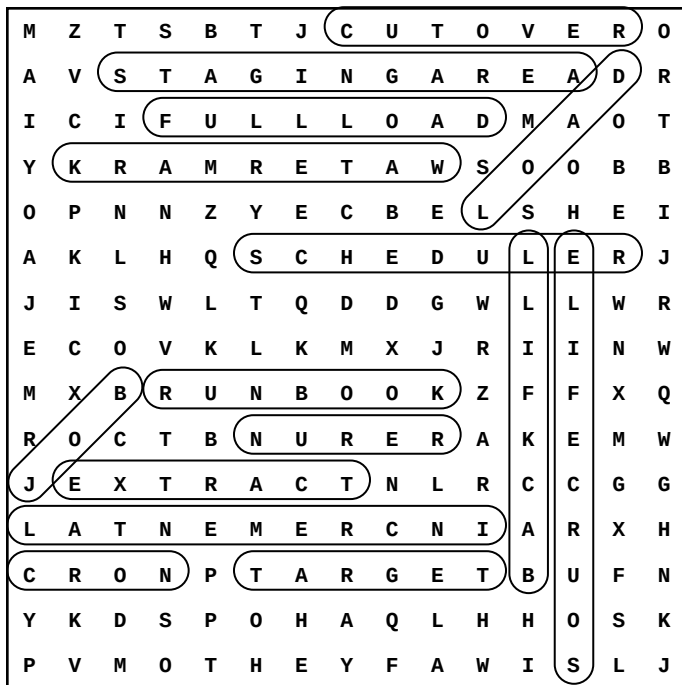
Solution 17



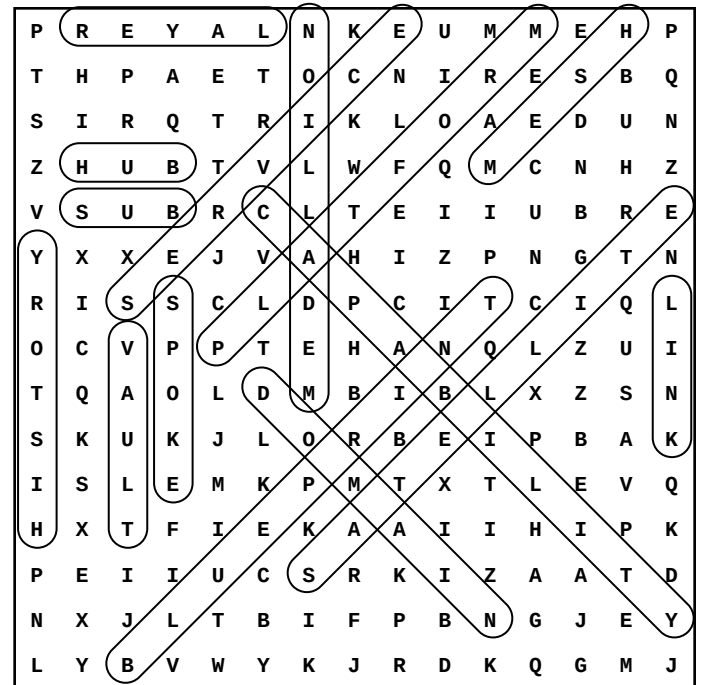
Solution 18



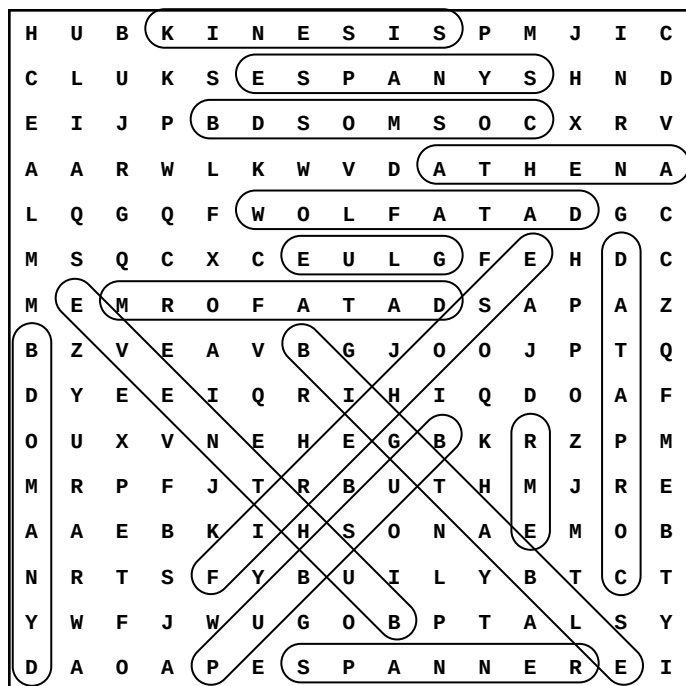
Solution 19



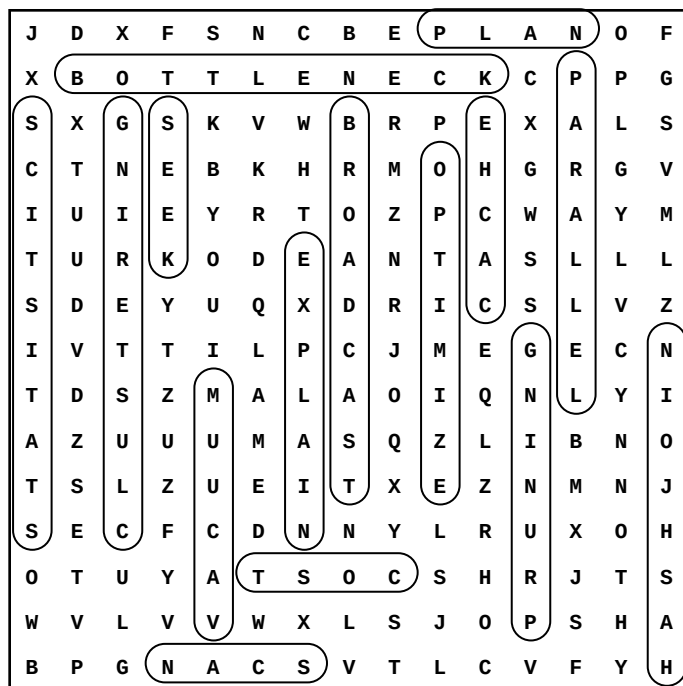
Solution 20



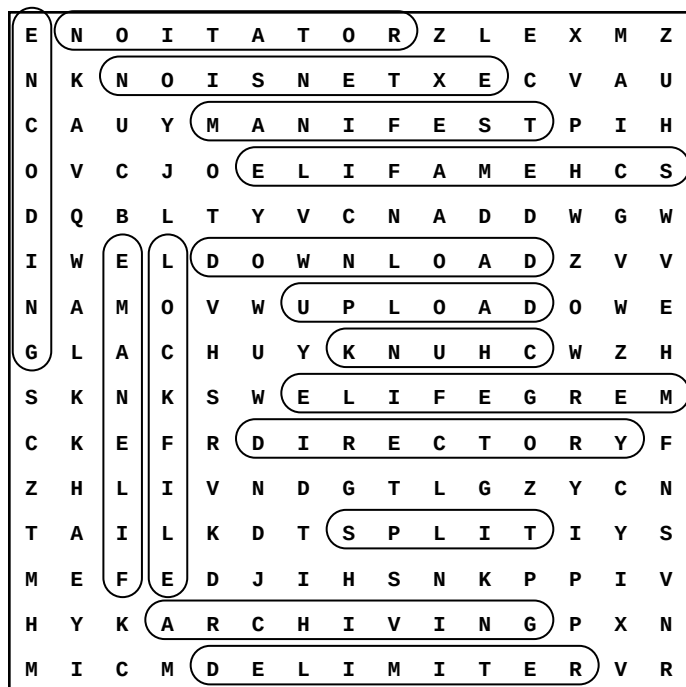
Solution 21



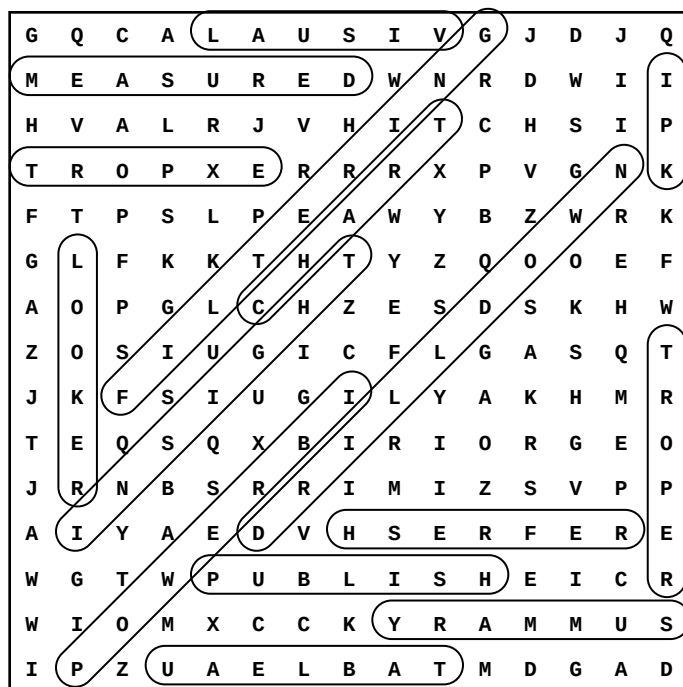
Solution 22



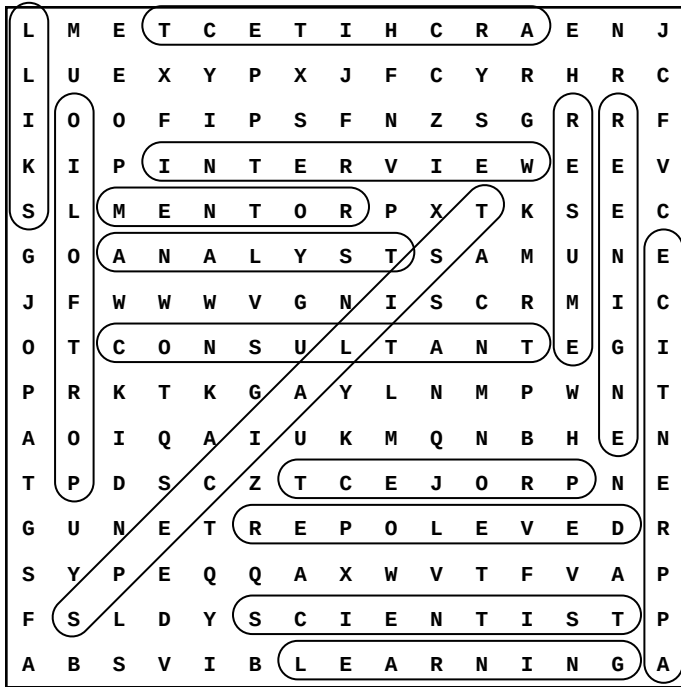
Solution 23



Solution 24



Solution 25



About the Author

Rob Lewis is a UK-based data professional with a passion for technology, automation and lifelong learning.

This puzzle book was generated using Python and built around real data engineering concepts, tools and technologies used throughout the industry.

Thank you for supporting this project and taking the time to solve these puzzles.

To my wife and daughter: thank you for your love, patience and encouragement. None of this would have been possible without you.

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Volume 1

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Find source code, updates and future projects:

<https://github.com/rob834lewis>